

**RESEARCH PROJECT REPORT
ON**

**"ANALYZING HISTORICAL SALES TRENDS
TO PREDICT FUTURE SALES"**

BY

J. ANGEL BENIT VERONICA

TABLE OF CONTENTS

CHAPTER NO.	NAME OF THE CHAPTER	PAGE NO.
CHAPTER 1	Research Environment	1
CHAPTER 2	Review of Literature	13
CHAPTER 3	Research Problem	17
CHAPTER 4	Data Analysis	19
	<u>Conclusion, Limitations, Recommendations</u>	
	<u>Bibliography</u>	
	Remark:	

CHAPTER 1: RESEARCH ENVIRONMENT

Located strategically within DRDO Phase 1 at CV Raman Nagar, Bangalore, Vidya Collections stands as a beacon of excellence in the stationary retail sector. Renowned for its comprehensive range of premium office supplies, art materials, creative tools, and educational resources, Vidya Collections has garnered widespread acclaim as a leader in its field. The store caters to a diverse clientele, including professionals, students, artists, and educators, reflecting its versatility and broad appeal.

DRDO Phase 1 in CV Raman Nagar is not just any location; it is a dynamic hub renowned for its cutting-edge scientific research and technological innovation. As an integral part of the Defence Research and Development Organization (DRDO), this campus serves as a catalyst for advancing knowledge and fostering collaboration across various scientific disciplines, with a particular emphasis on defence technology and related sectors. The environment pulsates with innovation, making it an ideal setting for businesses like Vidya Collections that thrive on creativity and innovation.

The strategic positioning of Vidya Collections within this vibrant ecosystem offers more than just access to high-quality stationery products; it provides a platform for creativity, knowledge exchange, and professional development. Customers visiting the store not only find the tools they need but also immerse themselves in an atmosphere conducive to inspiration and growth. This synergy between a leading stationary retailer and a pioneering scientific community creates a unique and enriching experience for all stakeholders involved.

Over the years, Vidya Collections has solidified its position as the preferred choice for stationary needs in DRDO Phase 1, Bangalore. Their unwavering commitment to upholding the highest standards of product quality, innovation, and professionalism has earned the trust and loyalty of their valued customers within this thriving scientific community.

Moreover, Vidya Collections is more than just a stationary store offering high-quality supplies like books and pens.

It understands the changing needs of its customers, especially in a dynamic environment like DRDO Phase 1. This involves regularly updating its inventory with the latest in office supplies and essentials, ensuring customers have access to top-notch tools for their work and studies. Vidya Collections' commitment to quality and customer satisfaction makes it the preferred destination for all stationary needs in DRDO Phase 1 and surrounding areas.

1.1. ASSUMPTIONS:

Data Representation:

The Excel document's three-month sales data is seen as representative of Vidya Collections' usual sales trends and customer interactions. Subsequent analyses and predictions are based on this assumption. However, it's crucial to acknowledge that the data might not encompass the influence of marketing campaigns, seasonal fluctuations, or external economic factors. These variables could significantly alter consumer behaviour and sales patterns, potentially affecting the reliability of the analyses and forecasts derived from the existing data. Understanding these limitations is key to making informed decisions and interpreting the results accurately in the context of broader market dynamics.

Economic Stability:

At the time of data collection, it is presumed that the state of the economy reflects Bangalore's overall environment. This includes constant purchasing power, steady consumer confidence, and the lack of major economic shocks. Accurately contextualizing the sales data and deriving significant insights depend on these kinds of assumptions. But it's important to understand that the state of the economy might change, which could influence how consumers behave and how the market works. Therefore, continuous monitoring and examination of economic trends remain essential for strong analysis and forecasting, even when the current data may reflect a particular economic setting. Making educated business judgments and comprehending the limitations of the data are made easier with this knowledge.

Market Trends:

Based on the presumption that market patterns observed during the data period will endure and have an ongoing influence on sales soon, the analysis was conducted. This includes both new trends, such as increased demand for eco-friendly products, and seasonal variations in sales, such as spikes during back-to-school seasons. The projection seeks to represent prevailing consumer preferences and market dynamics by integrating these trends into the research. However, as trends and consumer behaviour can change over time, it's imperative to keep an eye out for any changes in these areas. The analysis's accuracy and relevance in directing corporate planning and strategic decision-making are ensured by frequent updates.

Competitive Landscape:

The analysis makes the assumption that there are no substantial new competitors or disruptive changes in the tactics of existing ones inside DRDO Campus, Phase 1, CV Raman Nagar, which would materially affect Vidya Collections' position in the market. Accurately assessing Vidya Collections' performance and market share requires making this assumption. It's crucial to recognize, though, that competitive landscapes can change quickly as new firms enter the market or as established rivals employ cutting-edge tactics. Maintaining Vidya Collections' competitive edge and long-term performance requires constant observation of the market environment in order to adjust and react to potential changes.

Customer behaviour:

The research assumes that, barring an influence from outside variables not included in the data, consumer behaviour as seen in the data—including spending patterns, visit frequency, and purchase preferences—will stay consistent over the course of the forecast period. This presumption offers a starting point for forecasting future sales trends and formulating strategic choices. It's crucial to understand, though, that a variety of outside variables, such alterations in the economy, cultural norms, or new market trends, can have an impact on consumer behaviour. In order to ensure that Vidya Collections' projections and strategies remain aligned with changing consumer tastes and market realities, it is imperative that these external factors be regularly monitored.

Operational Stability:

The study proceeds under the premise that Vidya Collections' operating procedures which include pricing strategies, inventory management, and customer service standards continue to function normally and don't undergo any notable internal disturbances or changes. This presumption serves as the basis for examining sales data and making precise predictions about future performance. But it's important to recognize that internal factors—like management changes, problems with the supply chain, or adjustments to organizational priorities—can have an impact on operations. In order to adjust and maximize operational efficiency, flawless client experiences and long-term company success for Vidya Collections depend on constant monitoring and review of internal processes.

Technology Adoption:

Within the projected period, it is assumed that digitalization trends and technical improvements impacting Bangalore's stationery business will progress gradually instead of abruptly changing

the industry drastically. This presumption is in line with the present rate of technological adoption in the tech-savvy atmosphere of the region, where new ideas usually become part of the system gradually rather than generating abrupt upheavals. But it's important to keep an eye out for new technology and how they can change the way businesses operate. Vidya Collections maintains its competitiveness and aligns itself with changing client demands for digital solutions and experiences in the stationery sector through constant monitoring and adaption to technological advancements.

Regulatory Environment:

During the projection period, it is anticipated that there won't be any significant policy or regulatory changes that would have an immediate impact on Vidya Collections' ability to conduct business, set prices, or fulfil tax responsibilities. This presumption offers a robust legal framework within which to examine sales data and project future results. Nonetheless, it's critical to keep up with any regulatory changes that can have an impact on the business environment. For Vidya Collections, compliance, risk mitigation, and operational continuity are ensured by proactive monitoring and being ready to adjust to changes in regulations. This protects against unanticipated effects of regulations on pricing policies, tax obligations, and business operations.

1.2. PREMISES:

Data-Driven Approach:

Based on a data-driven methodology, this analysis makes use of historical sales data from November 2023 to January 2024. Finding patterns, trends, and insights that can be used to forecast future sales success is the aim. We can determine important variables impacting sales, like seasonal fluctuations, consumer preferences, and market trends, by evaluating this data. We can create reliable forecasting models and make wise business judgments with the help of this knowledge. Though past data offers insightful information, it's crucial to remember that unanticipated occurrences and outside variables may also have an impact on future sales success. Therefore, to achieve accurate sales predictions and optimize business outcomes, a combination of data analysis, market research, and strategic flexibility is essential.

Holistic Data Inclusion:

The analysis premise calls for taking into account a wide range of data items, such as net sales numbers, client information, invoice dates, and tax details. Through the integration of many

data pieces, the analysis aims to obtain full insights into the dynamics of sales. This method makes it possible to thoroughly examine a number of factors that affect sales success, including price tactics, seasonal trends, consumer behaviour, and tax consequences. Leveraging this vast array of data also makes it easier to produce more precise sales projections and strategic suggestions by enabling a more nuanced view of market dynamics. Such thorough data aspects improve the analysis's depth and dependability, enabling educated decision-making and maximizing commercial results for Vidya Collections.

Visualization and Trend Analysis:

Visualizing sales trends over the designated period is made easier by using Tableau for trend analysis. This idea emphasizes how important visual aids are for spotting sales trends, the impacts of seasonality, and any abnormalities. Analysts may produce interactive charts, graphs, and dashboards that offer a simple and clear perspective of sales data by utilizing Tableau's visualization features. This visual method improves comprehension and makes it easier for stakeholders to spot trends, patterns, and anomalies. Furthermore, Tableau visualizations may improve data-driven decision-making and strategic planning by revealing hidden insights and relationships within the data. All things considered, using Tableau for trend analysis improves the analytical procedure and enables businesses such as Vidya Collections to extract meaningful insights from their sales data.

Statistical Forecasting Using ARIMA:

Using Google Colab's ARIMA (AutoRegressive Integrated Moving Average) function gives the sales forecasting process more scientific rigor. This assumption acknowledges the value of time-series analysis in identifying underlying trends, seasonal fluctuations, and improving the accuracy of future sales forecasts. More accurate forecasts of future sales performance are made possible by analysts' use of ARIMA models, which take autocorrelation, trend, and seasonality in the sales data into consideration. To further improve the forecasting process, Google Colab offers an adaptable platform for data exploration, model creation, and result visualization. The integration of Google Colab with ARIMA enables analysts to derive valuable insights from sales data and make well-informed decisions that propel Vidya Collections' business expansion.

Historical Patterns as Predictors:

The idea is to forecast future sales success by utilizing past sales trends. The study attempts to generate well-informed forecasts for future sales periods by recognizing and comprehending historical patterns, such as revenue spikes during specific months or customer behavior trends. This method makes use of the insightful knowledge gleaned from past data to predict future shifts in client preferences, seasonal changes, and sales. Analysts may create forecasts and strategy plans for Vidya Collections that are more accurate by identifying and utilizing these trends into the forecasting model. This data-driven strategy makes sure that choices are supported by actual facts, which improves the business's capacity to precisely match client demand, manage inventories, and optimize marketing tactics.

Assumption of Continuity:

The assumption of continuity, which holds that past sales trends would continue to affect future sales absent materially different circumstances, is a fundamental tenet that underpins the forecasting process. This idea serves as the cornerstone for expanding on current sales patterns and applying them to project future sales figures. Assuming continuity allows analysts to use past data to find seasonal trends, reoccurring patterns, and consumer behaviour. This helps Vidya Collections make more educated decisions and more accurate forecasts. To ensure the forecasting model's dependability and relevance over time, it is crucial to stay alert to any external influences, such as changes in the economy, industry, or unanticipated events, that could upset these patterns.

Strategic Decision Support:

The ultimate goal of this research is to assist Vidya Collections in making strategic decisions. The analysis attempts to help with inventory planning, marketing tactics, resource allocation, and overall business growth activities by providing an accurate forecast of future sales performance. Optimizing operations, maximizing profitability, and boosting competitiveness in the market all depend on this strategic decision assistance. Vidya Collections can efficiently connect business plans with customer demand, anticipate market trends, and make educated decisions by utilizing data-driven insights and forecasts. In addition to increasing operational effectiveness, this proactive strategy helps Vidya Collections succeed in the long run in the cutthroat and fast-paced stationery market.

1.3. BOUNDARIES:

Data Scope:

The study is limited by the sales data that is now available, which includes invoice details, client information, tax breakdowns, and net sales amounts, and covers the period from November 2023 to January 2024. This particular time frame and the data columns it contains will be the only sources of information used for any analysis or predictions. This restriction highlights how crucial it is to operate within the parameters of the available data in order to produce insightful analysis and precise forecasts for Vidya Collections' sales success. Although the data is complete for the designated time period, it is important to recognize that changes or outside variables may affect sales trends and results after this point. As a result, the forecasts and analysis will be especially designed to take into account the data that is now accessible and how it will directly affect Vidya Collections' business strategies.

Non-Financial Factors:

Only financial factors will be included in the analysis; non-financial factors such as staff performance, customer happiness, brand perception, and operational issues will not be included unless they have a direct bearing on sales trends or are clearly visible in the data that is currently available. This methodology guarantees a focused and data-driven examination that is especially designed to comprehend and project Vidya Collections' sales performance. Financial measurements and their effect on sales dynamics within the given timeframe are given priority in this research, even if non-financial aspects could also be significant for overall business management. A more accurate and useful evaluation of sales patterns and forecasting accuracy is made possible by this focus.

External Market Factors:

While taking into account past sales patterns, the research does not account for unexpected occurrences, economic downturns, regulatory changes, or abrupt changes in the external market that could have a significant impact on sales beyond the studied historical period. This constraint emphasizes the necessity of continuous observation and adjustment to evolving market circumstances beyond the examined period. Even while previous data offers insightful information, outside variables might have a big impact on sales performance. In order to minimize risks and seize opportunities in the constantly changing market landscape, Vidya Collections must therefore continue to be flexible and sensitive to new trends and events.

Geographical Limitation:

Only sales data from Vidya Collections, located at the DRDO Shopping Complex in Bangalore, 560093, is included in the analysis. It does not account for geographical differences in consumer behaviour outside of this particular location, nor does it incorporate sales data from other branches or locations. This restriction emphasizes how crucial it is to place the study in the context of the available data and the dynamics unique to a given place. Although the analysis's conclusions are applicable to Vidya Collections at the designated location, further information would be needed to extrapolate the findings to other branches or areas, taking into account the peculiarities of each local market and the tastes of its customers.

Product Focus:

The research concentrates on analyzing sales of goods and services within the parameters of the provided data columns. It does not extend to analyzing specific product categories, pricing strategies, or market segments beyond what is reflected in the sales data. This focus ensures that the analysis remains aligned with the available information and aims to derive insights directly related to sales performance for Vidya Collections. While product categories, pricing strategies, and market segments are integral aspects of business strategy, their detailed analysis would necessitate additional data and a more comprehensive study tailored to each specific aspect of Vidya Collections' operations and market positioning.

Operational Constraints:

The analysis does not delve into operational challenges, supply chain issues, inventory management strategies, or internal processes unless they directly affect sales performance and are apparent in the provided data. This approach maintains a focused examination on factors directly influencing sales trends and forecasts for Vidya Collections. While operational aspects are crucial for overall business management, their inclusion in this analysis is contingent upon their observable impact on sales metrics within the specified timeframe and data columns. This targeted approach ensures a clear and actionable assessment of sales dynamics and forecasting accuracy, enabling strategic decision-making to drive sales growth effectively.

Forecasting Limitations:

While forecasting future sales using ARIMA is a component of the research, it's acknowledged that forecasting accuracy is contingent upon the quality and completeness of historical data, along with the assumptions made in the forecasting model. This recognition emphasizes the importance of robust data collection, data cleaning, and model validation processes to enhance

the reliability of the forecasts. Additionally, the transparency of assumptions and potential limitations of the forecasting model are crucial for interpreting forecast results accurately and making informed decisions based on the forecasted sales projections for Vidya Collections.

Single-Point Analysis:

The analysis provides a snapshot based on the available three months' data and does not consider longer-term trends, cyclical patterns, or seasonality effects beyond the observed period, unless they are explicitly evident in the data and relevant to the forecasting model. This approach ensures that the analysis remains focused on capturing immediate sales dynamics and making short-term predictions based on the provided data columns. While acknowledging the potential influence of longer-term trends or seasonal patterns, their inclusion in the analysis would require additional historical data and a more comprehensive modeling approach tailored to capturing such trends accurately for Vidya Collections.

1.4. SOCIO-ECONOMIC ENVIRONMENT:

Location Influence:

Vidya Collections operates within the socio-economic landscape of DRDO Shopping Complex in Bangalore – 560093. This location is distinguished by its proximity to educational institutions, research facilities, and residential areas, all of which significantly impact consumer behaviour and demand for stationery products. The presence of students, faculty members, researchers, and local residents contributes to a steady demand for various stationery items such as notebooks, pens, art supplies, and office essentials. Understanding the unique characteristics of this location is essential for Vidya Collections to tailor its product offerings, marketing strategies, and inventory management to meet the specific needs and preferences of its target customers within this socio-economic context.

Educational Hub:

Bangalore, often referred to as India's Silicon Valley, features a robust educational ecosystem with numerous schools, colleges, and universities. This thriving environment fuels the demand for stationery items among a diverse demographic, including students, teachers, researchers, and academic professionals. The demand for stationery products typically peaks during academic periods, such as the beginning of school semesters, exam seasons, and project submission deadlines. Understanding and catering to this dynamic demand is crucial for Vidya Collections to capitalize on the opportunities presented by Bangalore's educational landscape.

Tailoring marketing efforts, product offerings, and inventory management strategies to align with these peak periods can help Vidya Collections maximize sales and enhance customer satisfaction within this niche market segment.

Tech-Savvy Demographics:

Bangalore's population comprises a substantial segment of tech-savvy individuals employed in the IT sector and related industries. These demographics typically favour digital tools for various tasks but may still have a preference for traditional stationery items, especially for notetaking, planning, and creative work. This dual preference influences purchasing patterns, creating a unique market dynamic where there is a demand for both digital and traditional stationery products. Vidya Collections can capitalize on this trend by offering a diverse range of products that cater to the needs of tech-savvy professionals while also providing high-quality traditional stationery options. Understanding and adapting to the purchasing behaviours of this demographic can help Vidya Collections effectively serve its customer base and stay competitive in Bangalore's tech-driven market.

Income Levels:

The socio-economic environment in Bangalore's DRDO Shopping Complex encompasses varying income levels, ranging from students with limited budgets to professionals with higher disposable incomes. Understanding these income dynamics is crucial for Vidya Collections' pricing strategies, product offerings, and effective targeting of different customer segments.

For students and individuals with limited budgets, offering affordable yet quality stationery options can be appealing. This could include budget-friendly notebooks, pens, and other essentials that meet their academic needs without straining their finances. On the other hand, professionals and higher-income customers may be willing to invest in premium stationery products, such as luxury pens, designer notebooks, or personalized stationery sets.

By segmenting the market based on income levels and tailoring pricing strategies and product offerings accordingly, Vidya Collections can effectively cater to the diverse needs and preferences of its customer base. This approach can also help optimize revenue generation and enhance customer satisfaction within each income segment.

Cultural Influences:

Bangalore's diverse population and cosmopolitan culture significantly impact consumer preferences in the stationery market. There's a noticeable trend towards eco-consciousness, leading to increased demand for sustainable stationery options like products made from recycled materials or eco-friendly packaging. Alongside this, there's a growing preference for premium-quality stationery among affluent segments or those valuing luxury items, such as designer stationery sets or high-end pens.

Cultural events also influence buying behaviours, with consumers often purchasing themed stationery items for festivals or special occasions. Understanding and adapting to these trends are vital for Vidya Collections to effectively meet the diverse preferences of Bangalore's consumer base, aligning product offerings, pricing strategies, and marketing efforts accordingly.

Technological Advances:

Bangalore's tech hub status shapes consumer expectations for convenience, seamless online shopping experiences, and diverse digital payment options. Residents, accustomed to digital advancements, seek user-friendly interfaces, efficient checkout processes, and secure digital payments when purchasing stationery items. Vidya Collections can align with these expectations by optimizing its online platform for mobile use, providing intuitive navigation, and offering a range of secure payment methods. Adapting to these tech-driven consumer preferences can enhance customer satisfaction and competitiveness in Bangalore's market.

Market Competition:

The socio-economic environment includes competition from traditional brick-and-mortar stores within DRDO Shopping Complex and online platforms offering stationery products. Understanding competitive strategies, pricing dynamics, and customer loyalty is essential for market positioning. Vidya Collections must analyze competitor pricing strategies, promotions, and customer retention tactics to develop effective pricing strategies and promotional campaigns. Building customer loyalty through superior service, product quality, and personalized offerings can help differentiate Vidya Collections and enhance its market position amidst both physical and digital competitors.

Regulatory Framework:

Bangalore's regulatory environment, encompassing tax policies, licensing requirements, and compliance standards, can significantly impact Vidya Collections' operations, pricing

structures, and overall business viability. Changes in tax rates or regulatory requirements can directly affect the company's financial health and operational efficiency. Staying informed about regulatory updates and complying with legal standards are crucial for strategic decision-making and maintaining a competitive edge. By actively monitoring regulatory changes and adapting strategies accordingly, Vidya Collections can mitigate risks, ensure compliance, and optimize business operations for long-term success in Bangalore's dynamic market landscape.

Economic Stability:

Bangalore's economic stability, including factors like GDP growth, employment rates, inflation, and consumer confidence, significantly shapes purchasing power and consumer spending in the stationery industry. A thriving economy with strong GDP growth and low unemployment fosters consumer confidence and higher spending, positively impacting stationery sales. Conversely, economic downturns or inflationary pressures can curb consumer spending, affecting market trends. Understanding and monitoring these economic indicators are crucial for Vidya Collections to adjust pricing, offerings, and marketing strategies accordingly, ensuring resilience and growth within Bangalore's dynamic stationery market.

CHAPTER 2: REVIEW OF LITERATURE

Halder, P. K., Biswas, S. K., Islam, M. T., & Beg, M. R. A. (2015): This paper explores a case study that focuses on error determinants and forecasting method selection in the Bangladeshi jute product manufacturing industry. A number of forecasting methods were assessed using demand data from Akij Jute Mills for the years 2010–2013. Simple moving averages fell between 27 and 35, exponential smoothing between 28 and 30, least squares linear regression between 25.83 and the MAPE of 26.55 for the Naïve forecast. With a MAPE of 16.29, the weighted moving average stood out as performing better. This study highlights how important it is to choose the right forecasting method, provides Bangladeshi manufacturers with a useful manual, highlights the usefulness of weighted moving averages for predicting demand in the jute industry, and may even be used as a template for forecasting efforts by other businesses.

Bansal, A., Elkady, G., & Singh, R. B. (2022): This paper examines the field of stock market forecasting and emphasizes the importance of this field for GDP and economic growth. In addition to serving as canisters of commerce and human capital, stock markets also serve as a symbol of the volatility of profit margins. Precisely predicting the performance of the stock market can result in significant profits during successful times and reduce losses during downturns. "Stock market prediction" is speculating about the future value of financial instruments and stocks of exchange-traded companies, frequently using news sentiment to improve accuracy. Prediction models such as LSTM, Prophet, and ARIMA are integrated with quantitative analysis, behavioural economics, and market psychology to increase forecasting accuracy. The study highlights the significance of assessing market trends and suggests an integrated model that performs better than separate models, proving its usefulness in lowering error margins and forecasting future stock values.

Udokwu, C., Brandtner, P., Darbanian, F., & Falatouri, T. (2022): This paper explores the topic of Point of Sale (PoS) retail and how important it is to everyday customer interactions. Even with the growth of e-commerce, physical stores are still important since they provide not just product sales but also cutting-edge services like omnichannel assistance and leisure areas. Because retail processes create large amounts of statistics, data analytics becomes a powerful tool for improving performance through the optimization of product availability and sales prediction. Low-quality projections result from the difficulties in projecting sales across a variety of products and retail locations. In order to create a single model for sales prediction,

the article suggests clustering stores with comparable properties by utilizing machine learning techniques. In order to improve sales estimates in physical retail, it also addresses the paucity of research on systematic store grouping, identifying critical factors and assessing clustering and prediction algorithms. The findings emphasize the significance of variables such as pricing dynamics, location features, market share, competition, and foot traffic, with K-Means clustering and random forest prediction showing the best outcomes. Notwithstanding the fact that the study provides insightful information, its shortcomings include single-case selection and regional data bias. As a result, more research is necessary to address irregular sales patterns brought on by events such as stock-outs and promotions and to enhance data pretreatment.

Viedienieiev, V. A., & Piskunova, O. V. (2021): Using barley prices as an example, this research explores the efficacy of machine learning algorithms in agricultural commodity price forecasting. It compares deep learning algorithms and conventional forecasting techniques, looking at which is more appropriate for usage in business and government. Time series forecasting techniques used in the study include direct-recursive hybrid convolutional neural networks, Fb Prophet, recursive multi-step, and linear regression. The findings show that while neural networks and conventional approaches both perform better than naive predictions, traditional models are often more successful and resource-efficient. Nonetheless, neural networks show great proficiency in recognizing a variety of patterns, indicating possible advantages when merged with classical techniques through ensemble approaches. Research, agricultural enterprises, related industries, and public planning and control activities can all benefit from the study's conclusions. Future directions for research could involve investigating ensemble models and multifactor forecasting that incorporates additional data sources such as GDP, exports, yields, and international news.

Arunraj, N. S., Ahrens, D., Fernandes, M., & Müller, M. (2014): This paper examines how German retailers are using a SARIMAX model to decrease food waste and enhance sales forecasting. SARIMAX outperforms SARIMA by 5% and provides superior accuracy and outlier explanation by factoring in variables such as promotions and holidays. This improvement has the potential to improve forecasts and save waste. The study emphasizes how important it is to take a variety of elements into account when forecasting, particularly in the retail sale of perishable goods, where an accurate prediction can have a big impact on waste reduction. In the end, these developments help retail operations by streamlining inventory management, and they also support sustainability initiatives by reducing food waste across the supply chain.

Ekpanithanpong, K., & Jitkuakul, P. (2019): The purpose of the study was to investigate how brand engagement affects decision-making, trust, and loyalty in a stationery store in Sakon, Nakhon Province. Strong relationships were found between brand engagement and decision-making, trust, and loyalty, especially among female customers aged 21 to 30 who earned between 15,000 and 25,000 Baht per month and had a bachelor's degree. Only loyalty had a substantial impact on decision-making ($\beta=0.86$), while trust and loyalty were strongly influenced by brand engagement ($\beta=0.60, 0.46$). The report makes recommendations for tactics to improve brand perception and consumer loyalty, like loyalty programs and social welfare projects. By combining brand engagement, trust, and loyalty, businesses may better understand their customers' habits, adapt to their requirements, improve convenience, encourage repurchasing, and build long-lasting connections. Plans for future study include qualitative investigations for brand development using questionnaires created by the researcher.

Moorthy, A. L., & Pant, A. (2012): Founded in 1958, the Defence Research and Development Organisation (DRDO) in India is a leading research and development organisation dedicated to developing advanced technologies for national security. With more than 53 labs and around 7000 scientists, DRDO is involved in a wide range of subjects, including biomedical sciences, weapons, and aeronautics. Because their job is dynamic, DRDO scientists need to have access to current worldwide research. By using consortiums to embrace e-resources, DRDO has saved a significant amount of money every year—more than Rs 5 crore. The utilization of resources and the continued efforts to extend consortium services demonstrate the good impact. In contrast to academic institutions, DRDO's emphasis on defense research calls for special methods of resource sharing and information access, which are successfully enabled via DESIDOC.

Hasugian, P. M., Halawa, F., & Kartighaiyab, D. R. (2022): This study highlights the crucial role of ATK (Office Stationery) in supporting operational activities at STMIK Pelita Nusantara. However, the lack of analysis on ATK purchase data poses a challenge. The researcher aims to extract valuable insights and predict ATK needs using the Frequent Pattern-Growth (FP-Growth) algorithm. Unlike the commonly used a priori algorithm, FP-Growth efficiently identifies frequent itemsets by creating an FP-Tree that minimizes database scans and speeds up the mining process. By employing association rules with FP-Growth, correlations between ATK needs and available supplies can be identified, aiding in effective inventory management and resource allocation.

Bugis, M., Kembauw, E., Sangadji, M., Djunaidi, F. G., Umanailo, M. C. B., Nur, H., & Tang, B. (2021): The importance of marketing in increasing market share and profitability through useful tactics and offerings is emphasized in this study. Businesses face fierce competition, so it's critical to comprehend consumer behaviour and the elements that influence purchases. Both internal corporate efforts and external influences, such as personal characteristics, income brackets, and lifestyle preferences, have an impact on the decision-making process. Focusing on satisfying customer needs and product quality is emphasized as a competitive advantage for businesses. Pricing, brand recognition, and product variety are some of the elements that affect consumer decisions in Namlea City, where there are a diversity of consumer demographics. In order to stay relevant in the market, companies like Store Ilmu Store strive to offer a wide range of products at competitive prices, as well as first-rate customer service, even in the face of obstacles like the Covid-19 pandemic that reduce purchasing capacity. Positive correlations are found when product completeness and pricing are analyzed in relation to consumer purchase decisions at the UD. Pemancar Ilmu Store in Namlea City. This indicates the importance of these aspects in influencing consumer behavior and boosting sales.

Suwarno, S., & Kusnadi, R. (2021, April): Happy Stationery, located in Komplek Tanjung Pantun, Batam, has trouble keeping track of inventory because of its unorganized and paper-based use of Excel. A Warehouse Management System (WMS) was created utilizing the Software Development Life Cycle (SDLC) technique in order to address this. After data was gathered through observations and interviews, a web-based solution that improves inventory management and satisfies user needs was created. Nevertheless, observations and interviews show that Happy Stationery's operating mechanisms are still deficient. Streamlining sales, purchasing, and inventories with a digital system boosts productivity and enhances customer support. Additionally, the web-based system has warehouse management features that enable more precise product location tracking and quicker customer response. Happy Stationery's operational effectiveness and customer happiness have significantly increased as a result of the transition from paper-based to digital systems.

CHAPTER 3: RESEARCH PROBLEM

3.1. PURPOSE OF RESEARCH:

The research project is centered on conducting a detailed trend analysis and sales forecast using three months of historical sales data. The trend analysis, performed using Tableau, will help uncover patterns and seasonality in sales data, providing valuable insights into sales trends over time. Additionally, the sales forecast will be generated on Google Colab. The main objective of this research is to accurately predict future sales based on historical data trends. By leveraging data visualization and statistical modeling, this study aims to enhance sales forecasting accuracy.

3.2. RESEARCH DESIGN:

The research design revolves around conducting a detailed analysis of historical sales data sourced directly from the store, with a specific focus on trend analysis and sales forecasting. The primary objective is to leverage this data to gain insights into sales patterns and predict future sales accurately. The research will involve data collection, cleaning, and preparation to ensure data quality. Trend analysis will be performed using Tableau, allowing for visual exploration of sales trends, patterns, and seasonality. Additionally, the ARIMA model will be implemented in Google Colab for sales forecasting, utilizing past sales data to predict future performance. The research design also encompasses validation methods to assess the accuracy and reliability of the sales forecast. Ethical considerations regarding data privacy and confidentiality will be upheld throughout the study. The expected outcome is a comprehensive understanding of sales trends and a reliable sales forecast that can guide strategic decision-making for the business.

3.3. SOURCE OF DATA:

The principal data source for this study is secondary data that was acquired from a respectable retail institution as an Excel file. Numerous specific details about sales transactions are covered by the Excel sheet, such as invoice dates, invoice numbers, client names, taxable values, SGST and CGST amounts, exempted sales, round-off values, and net sales statistics. Throughout a three-month period, this secondary data provides a fundamental resource for doing detailed trend analysis and accurate sales forecasting. Strict data governance guidelines will be followed in the implementation of data cleaning, validation, and preparation procedures to guarantee the correctness, consistency, and integrity of this data. The research team will be able to obtain

actionable insights and provide data-driven suggestions for improving business performance and strategic decision-making about inventory management, resource allocation, and sales optimization techniques by utilizing this substantial secondary dataset.

3.4. OBJECTIVE:

Predicting sales performance based on historical sales.

Trend analysis will be performed using Tableau, allowing for visual exploration of sales trends, patterns, and seasonality. Additionally, the ARIMA model will be implemented in Google Colab for sales forecasting, utilizing past sales data to predict future performance.

CHAPTER 4: DATA ANALYSIS

4.1. RESEARCH METHODOLOGY

The research methodology for this study involves a systematic approach to analysing historical sales data, conducting trend analysis, and performing sales forecasting. The primary data source is an Excel sheet provided by the shopkeeper, containing detailed information on sales transactions such as invoice dates, numbers, customer names, taxable values, SGST and CGST amounts, exempted sales, round-off values, and net sales. The data undergoes rigorous cleaning and preparation processes to ensure accuracy, including handling missing values and outliers. Trend analysis is conducted using Tableau, focusing on visualizing sales patterns, identifying seasonality, detecting anomalies, and exploring potential influencing factors. For sales forecasting, the ARIMA model is implemented in Google Colab, trained on historical sales data to predict future performance based on established trends. Validation of the sales forecast is performed using statistical metrics and comparison with actual sales data to ensure accuracy and reliability. Ethical considerations regarding data privacy and integrity are strictly adhered to throughout the research process. The analysis phase involves interpreting the results to derive actionable insights, such as identifying key sales trends and providing recommendations for strategic decision-making.

ARIMA MODEL:

The ARIMA (AutoRegressive Integrated Moving Average) model is a powerful tool in time series analysis, widely used for forecasting future values based on past data patterns. Its name reflects its key components: AutoRegressive (AR), Integrated (I), and Moving Average (MA).

AutoRegressive (AR):

The "AR" part of ARIMA signifies that the model uses the relationship between an observation and a certain number of lagged observations (i.e., past values) to predict future values. It assumes that the future value of a series is linearly dependent on its previous values.

Integrated (I):

The "I" component represents differencing, which involves transforming a non-stationary time series into a stationary one. Stationarity is crucial because many time series models, including ARIMA, work under the assumption that the statistical properties of the series, such as mean and variance, remain constant over time.

Moving Average (MA):

The "MA" part refers to the moving average component, which models the relationship between an observation and a residual error from a moving average model applied to lagged observations.

Key Steps in Using ARIMA:

Data Analysis and Preprocessing: Understand the data, check for trends, seasonality, and stationarity. If the series is not stationary, perform differencing until stationarity is achieved.

Parameter Selection: Determine the order of differencing (d), the number of autoregressive terms (p), and the number of moving average terms (q) based on autocorrelation and partial autocorrelation plots.

Model Fitting: Fit the ARIMA model to the preprocessed data using the chosen parameters.

Model Evaluation: Evaluate the model's performance using statistical measures such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). Additionally, diagnostic plots can help assess the model's goodness of fit.

Forecasting: Once the model is validated, use it to forecast future values of the time series.

Applications of ARIMA:

Finance: ARIMA models are extensively used in financial forecasting, such as predicting stock prices, currency exchange rates, and interest rates.

Economics: They are applied in economic modeling for GDP forecasting, inflation prediction, and consumer spending analysis.

Weather Forecasting: ARIMA models play a role in weather prediction by analyzing historical weather data to forecast future climate conditions.

Sales and Demand Forecasting: Businesses use ARIMA to forecast sales, demand for products, and inventory levels.

Healthcare: In healthcare, ARIMA models can be utilized for predicting patient admission rates, disease outbreaks, and medical resource allocation.

Challenges and Considerations:

ARIMA assumes linearity and stationarity, which may not hold true for all time series data.

Proper model selection and tuning of parameters are crucial for accurate forecasts.

It may not perform well with highly volatile or irregular data patterns.

4.2. DATA ANALYSIS:

Data:

This data contains information about sales invoices, from Vidya Collections (stationary store) at DRDO Shopping Complex. The data initially consisted of invoices across November, December, and January.

The principal data source for this study is secondary data that was acquired from a respectable retail institution as an Excel file. Numerous specific details about sales transactions are covered by the Excel sheet, such as invoice dates, invoice numbers, client names, taxable values, SGST and CGST amounts, exempted sales, round-off values, and net sales statistics. Throughout a three-month period, this secondary data provides a fundamental resource for doing detailed trend analysis and accurate sales forecasting. Strict data governance guidelines will be followed in the implementation of data cleaning, validation, and preparation procedures to guarantee the correctness, consistency, and integrity of this data. The research team will be able to obtain actionable insights and provide data-driven suggestions for improving business performance and strategic decision-making about inventory management, resource allocation, and sales optimization techniques by utilizing this substantial secondary dataset.

All the data was combined into a single excel spreadsheet.

INV DATE - Invoice Date: The date the invoice was issued.

INV NO - Invoice Number: Unique identifier for each invoice.

CUSTOMER NAME - These columns contain customer names associated with the invoices.

TAXABLE VALUE - The monetary value of the goods or services before taxes.

SGST - State Goods and Services Tax, a tax levied by the Indian government specific to the supplying state.

CGST - Central Goods and Services Tax, a tax levied by the Indian government applicable throughout the country.

EXEMPTED SALE - The amount of the sale exempt from sales tax.

ROUND OFF - The adjustment amount to ensure the total invoice amount has no decimals.

NET AMOUNT - The total amount payable by the customer after tax calculations and adjustments.

TAX VALUE - The total amount of sales tax applied to the invoice.

Trend Analysis:

The goal of this research's trend analysis is to detect, illustrate, and explain patterns, seasonality, and trends over a three-month period by thoroughly examining historical sales data using Tableau. Finding potential influencing causes, identifying underlying patterns or cycles, and comprehending the direction and size of sales changes are the main goals.

The research tries to emphasize sales patterns over time and provide insights into the overall sales performance trajectory through data visualization techniques like line charts, bar graphs, and trend lines. The research aims to obtain practical insights for strategic decision-making, such as maximizing inventory levels, organizing marketing campaigns, and spotting chances for sales growth, by examining past sales data.

An essential component of this research is trend analysis, which entails a detailed Tableau examination of past sales data. Finding, visualizing, and interpreting patterns, seasonality, and trends across a three-month span is the aim. Understanding the direction and size of sales fluctuations, spotting underlying patterns or cycles, and identifying potential influencing factors are the main goals of the analysis.

To illustrate sales patterns over time and offer insights into overall sales performance, data visualization techniques including line charts, bar graphs, and trend lines are employed. The goal of the research is to produce useful information for strategic decision-making, such as determining chances for sales growth, organizing marketing campaigns, and managing inventory levels.

The objective of this research's trend analysis is to detect, visualize, and understand patterns, seasonality, and trends over a three-month period by thoroughly examining historical sales data using Tableau. The analysis looks into identifying underlying patterns or cycles, determining the direction and size of sales changes, and locating probable influencing factors.

The research tries to highlight sales patterns over time and provide insights into the overall sales performance trajectory using data visualization tools including line charts, bar graphs,

and trend lines. The research attempts to produce practical insights for strategic decision-making, such as maximizing inventory levels, organizing marketing campaigns, and spotting chances for sales growth, by examining past sales data.

Sales Forecast:

This study used an advanced statistical method called the ARIMA (AutoRegressive Integrated Moving Average) model to predict future sales performance with high accuracy. The ARIMA model uses the moving average (MA) component to account for random fluctuations, the integrated (I) component to address trends and seasonality, and the auto-regressive (AR) component to capture the linear relationship between past and present sales. The historical sales data that was obtained from the store is the basis for this analysis.

The ARIMA model, which is implemented with Google Colab, is trained on past sales patterns in order to produce accurate forecasts of future sales trends. This modeling technique adds to a more reliable and accurate sales forecasting methodology by taking into account not just historical sales behaviour but also potential seasonality effects and trends.

This research relies heavily on the ARIMA (AutoRegressive Integrated Moving Average) model, which provides an accurate projection of future sales performance. With the help of the store's historical sales data, the ARIMA model combines the moving average (MA) component to manage sporadic fluctuations, the integrated (I) component to handle trends and seasonality, and the auto-regressive (AR) component to capture the linear relationship between past and present sales.

The ARIMA model, which is implemented using Google Colab, is trained on past sales patterns in order to produce accurate forecasts of future sales trends. This modeling technique produces a more reliable and accurate sales forecasting methodology by taking into account not just historical sales behaviour but also potential seasonality effects and trends.

By precisely projecting future sales performance, the ARIMA (AutoRegressive Integrated Moving Average) model improves the forecasting powers of the research. The model makes use of historical sales data and employs the moving average (MA) component to handle random fluctuations, the integrated (I) component to manage trends and seasonality, and the auto-regressive (AR) component to determine the relationship between past and present sales.

When the ARIMA model is used on Google Colab, it is rigorously trained on past sales patterns in order to produce accurate forecasts of future sales trends. This strategy contributes to a more

thorough and accurate sales forecasting process by taking into account prospective seasonality effects and trends in addition to prior sales behaviour.

This study makes extensive use of the ARIMA (AutoRegressive Integrated Moving Average) model, which offers a complex statistical framework for precisely projecting future sales performance. The auto-regressive (AR) component of the ARIMA model, which uses historical sales data from the store, addresses trends and seasonality, the integrated (I) component handles linear relationships between past and present sales, and the moving average (MA) component manages erratic fluctuations.

The ARIMA model, which is used with Google Colab, is trained extensively on past sales trends in order to generate accurate forecasts of future sales trends. This modeling technique adds to a more reliable and accurate sales forecasting methodology by taking into account not just historical sales behaviour but also potential seasonality effects and trends.

Trend Analysis on Tableau:

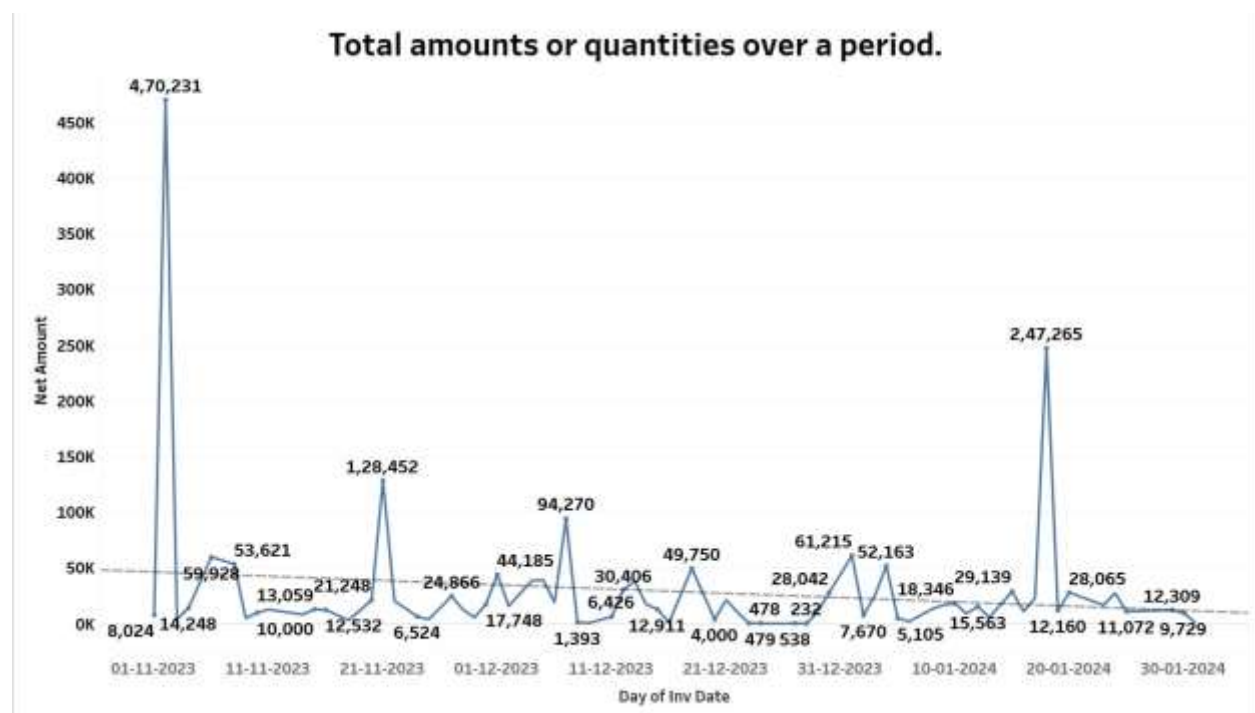


Figure 1: Total amounts or quantities over a period.

Interpretation:

- The x-axis represents “Invoice date” and the y-axis represents “Net amount”.

- Overall, net amount exhibits a positive trend from November 1, 2023 to January 30, 2024, with some fluctuations.
- A significant increase in net amount is observed in November, 2023, reaching a peak of nearly 470,231.
- A significant increase in net amount is observed on December 31, 2023, reaching a peak of nearly 470,231. This could be attributed to factors like year-end sales or invoicing practices.
- The graph also reveals a potential seasonal trend, with a steady rise in net amount leading up to December 31st, followed by a decrease in January. This may be associated with holiday sales patterns.

Sales Forecasting on Google Colab using ARIMA model:

Code:

```
from pandas import read_excel

import matplotlib.pyplot as plt

from statsmodels.tsa.arima.model import ARIMA

# Upload your data from the Files section

data = read_excel('DATA.xlsx') # Adjust file name if needed

# Convert the 'DATE' column to datetime format

data['DATE'] = pd.to_datetime(data['DATE'])

# Set 'DATE' as the index

data.set_index('DATE', inplace=True)

# Fit an ARIMA model (replace with your preferred model)

model = ARIMA(data['NET AMOUNT'], order=(1, 1, 1)) # Adjust ARIMA order as needed

model_fit = model.fit()

# Generate predicted values for the next month

future_days = pd.date_range(start=data.index[-1] + pd.Timedelta(days=1), periods=30)
```

```

predicted_sales = model_fit.forecast(steps=30)

# Plot the data and predicted sales

plt.figure(figsize=(12, 6))

plt.plot(data['NET AMOUNT'], label='Sales')

plt.plot(future_days, predicted_sales, label='Predicted Sales')

plt.legend()

plt.xlabel('Date')

plt.ylabel('Net Amount')

plt.title('Sales Forecast for Next Month with ARIMA Model')

plt.grid(True)

plt.show()

print("Predicted Sales for the Next Month:")

for date, sales in zip(future_days, predicted_sales):

    print(f'{date.strftime('%Y-%m-%d')}: {sales:.2f} ")

```

Graph Output:

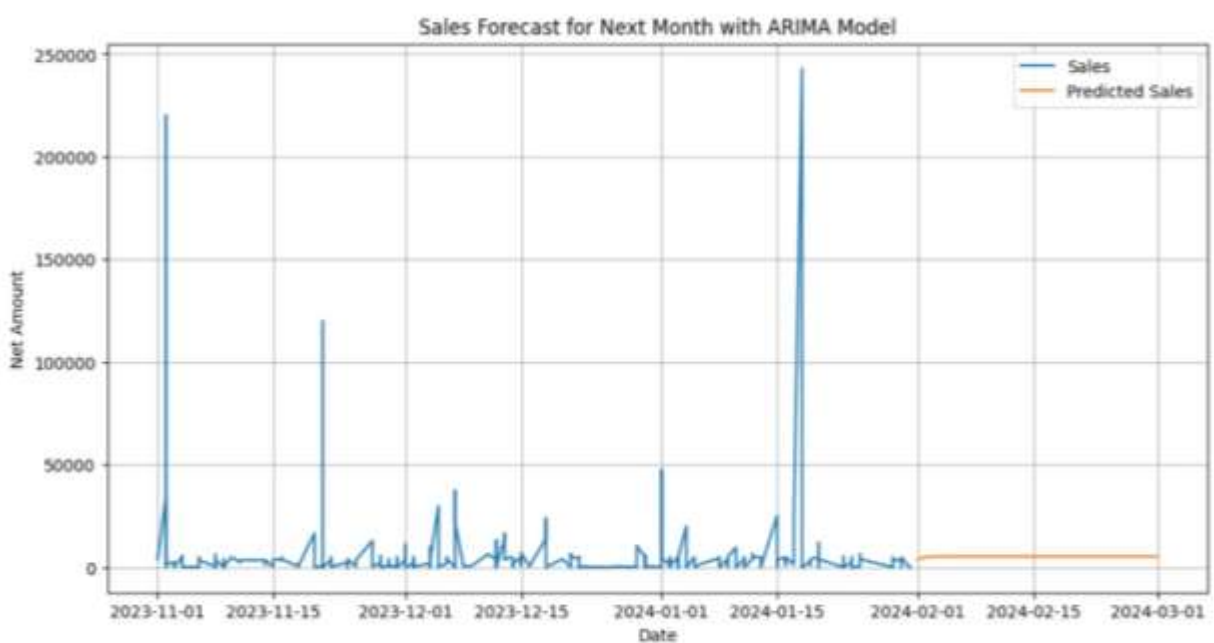


Figure 2: Sales Forecast for Next Month for Vidya Collections, DRDO, with ARIMA Model

Predicted output from code:

Predicted Sales for the Next Month:

2024-02-01: 4072.28

2024-02-02: 5038.53

2024-02-03: 5271.86

2024-02-04: 5328.21

2024-02-05: 5341.82

2024-02-06: 5345.10

2024-02-07: 5345.90

2024-02-08: 5346.09

2024-02-09: 5346.13

2024-02-10: 5346.15

2024-02-11: 5346.15

2024-02-12: 5346.15

2024-02-13: 5346.15

2024-02-14: 5346.15

2024-02-15: 5346.15

2024-02-16: 5346.15

2024-02-17: 5346.15

2024-02-18: 5346.15

2024-02-19: 5346.15

2024-02-20: 5346.15

2024-02-21: 5346.15

2024-02-22: 5346.15

2024-02-23: 5346.15

2024-02-24: 5346.15

2024-02-25: 5346.15

2024-02-26: 5346.15

2024-02-27: 5346.15

2024-02-28: 5346.15

2024-02-29: 5346.15

2024-03-01: 5346.15

Interpretation:

- This graph depicts a sales forecast for February 2024, generated using an ARIMA model.
- The orange line represents the predicted sales trend for the entire month of February.
- The blue line represents historical sales data, from November 2023 to February 2024.
- Specific values for each day's forecast are provided, ranging from 4,072.28 on February 1st to 5,346.15 on February 15th and remaining constant thereafter.

4.3. FINDINGS:

Flat Trend Analysis:

The analysis of the sales trend for February unveils a stable trajectory with a subtle increase in sales during the first half of the month, as illustrated by the orange line in the graphical representation. The forecasted figures highlight consistent patterns until February 15th, with projected sales hovering around 5,346.15 units, marking a phase of moderate growth. This observation underscores the importance of monitoring sales dynamics closely to identify underlying trends and capitalize on growth opportunities.

Delving deeper into the analysis, it becomes crucial to examine the factors contributing to this moderate growth phase. By dissecting sales data leading up to February 15th, it is possible to pinpoint specific drivers or market conditions that may have influenced this upward trend. This

level of granularity in the analysis allows for a more comprehensive understanding of sales dynamics, enabling businesses to make informed decisions and strategic adjustments.

Moreover, understanding market conditions and drivers behind sales fluctuations provides valuable insights for optimizing strategies and maximizing sales performance moving forward. It allows businesses to adapt their approach, such as adjusting inventory levels, fine-tuning marketing campaigns, or targeting specific customer segments more effectively. Ultimately, this detailed analysis of the February sales trend not only sheds light on current performance but also lays the groundwork for proactive and data-driven decision-making in the future.

Comparison with Historical Data:

The juxtaposition of forecasted sales against historical data, depicted by the blue line, provides a valuable opportunity to discern significant deviations from established trends. This comparative analysis is instrumental in identifying patterns, anomalies, and potential influencing factors that could impact sales performance in distinct ways.

By closely examining the relationship between forecasted and historical sales, businesses can gain insights into the accuracy of their forecasting models and the reliability of their historical data. Any notable deviations or discrepancies can be further investigated to understand the underlying causes, whether they stem from external market factors, internal operational changes, or other variables.

This analysis not only helps in refining forecasting methodologies but also informs strategic decision-making. By recognizing patterns and anomalies, businesses can proactively adjust their strategies, optimize resource allocation, and capitalize on emerging opportunities or address potential challenges. Ultimately, the juxtaposition of forecasted and historical sales data serves as a powerful tool for enhancing sales forecasting accuracy and driving informed business decisions.

Model Limitations and Considerations:

It's important to recognize the inherent limits of the ARIMA model, even though it offers a systematic framework for examining historical seasonality and trends. Expected sales trends may be disrupted by unforeseen events, market dynamics, or outside factors, deviating from the expected projection. Therefore, even while the model provides useful insights, making wise decisions requires a comprehensive grasp of the larger market landscape as well as the ability to modify strategy in response to changing circumstances.

Businesses must complement quantitative models like ARIMA with qualitative analysis and contextual understanding to navigate uncertainties effectively. This comprehensive approach enables organizations to anticipate potential disruptions, mitigate risks, and seize emerging opportunities, thereby enhancing their ability to make informed decisions and sustain long-term growth.

CONCLUSION

In conclusion, the analysis of sales patterns using the ARIMA model provides valuable insights into the expected progression of net amounts for February 2024, as depicted graphically from November 1, 2023, to January 30, 2024. During this period, a positive trend in net amounts is evident, with notable peaks and swings in November and December 2023. These fluctuations in net amounts can be attributed to various factors such as holiday sales patterns, year-end sales, and invoicing procedures, all of which are reflected in the graph, indicating potential seasonal tendencies aligned with typical consumer behavior.

The comparison of the projected sales trend (represented by the blue line) with historical data provides valuable insights into deviations from established patterns, allowing for the identification of trends, anomalies, and potential influencing factors affecting sales performance.

In summary, the sales trend projection for February 2024 suggests a period of moderate growth, with an expected average of 5,346.15 units sold by the 15th of February. This observation indicates a phase of stability and moderate growth in line with historical patterns and seasonal variations.

LIMITATIONS:

Limited Data Period:

The analysis is based on a relatively short period from November 1, 2023, to January 30, 2024. This limited timeframe may not capture long-term trends or seasonal variations that extend beyond this period.

Single Model Approach:

The sales forecasting relies solely on the ARIMA model. While ARIMA is a robust choice, using only one modeling technique may overlook alternative models or ensemble methods that could potentially provide more accurate forecasts, especially in complex and dynamic market environments.

Omitted Variables:

The analysis primarily focuses on sales data and model predictions without incorporating other relevant variables such as marketing expenditures, economic indicators, or competitor actions. Omitting these variables may limit the depth of understanding regarding the drivers of sales fluctuations and the overall market dynamics.

Assumed Seasonality:

The interpretation assumes seasonal trends based on historical data patterns, such as holiday sales. However, without explicit seasonal adjustment techniques or deeper statistical tests for seasonality, the observed trends may not accurately represent true seasonal effects.

Limited Forecast Horizon Evaluation:

While the forecasted sales for February 2024 are presented, there is limited evaluation or validation of the forecast accuracy against actual sales data for that period. Without such validation, the reliability of the forecasted figures may be uncertain.

Model Calibration and Parameter Sensitivity:

The ARIMA model's performance can be sensitive to the chosen parameters (e.g., order of differencing, AR and MA terms). Small variations in these parameters could lead to significantly different forecast outcomes. Robustness tests or sensitivity analyses could provide insights into the stability of the model's predictions.

External Events Impact:

The analysis acknowledges potential factors like holiday sales but does not explicitly consider the impact of unforeseen external events (e.g., supply chain disruptions, regulatory changes, economic shocks) that could significantly influence sales patterns and deviate from historical trends.

Sample Representation:

It's essential to consider whether the data sample used for analysis adequately represents the entire population or market segment of interest. Biases or skewness in the sample could affect the generalizability of the findings to the broader context.

Future Data Uncertainty:

The forecasted sales figures are presented with a high level of precision for the entire month of February 2024. However, inherent uncertainties in future data, such as demand fluctuations or unexpected market shifts, may challenge the accuracy of these precise forecasts.

RECOMMENDATIONS:

Refine Seasonal Sales Strategies:

Given the observed seasonal trends, especially the increase in net amount leading up to December 31st followed by a decrease in January, Vidya Collections can refine its seasonal sales strategies. This might include targeted promotions, special offers, or themed collections to capitalize on the holiday sales peak and maintain customer engagement in January.

Optimize Inventory Management:

Use the sales forecasting insights generated from the ARIMA model to optimize inventory levels. Adjust stock quantities based on predicted sales trends, ensuring that Vidya Collections has the right products available at the right times to meet customer demand without excess inventory.

Data-Driven Marketing Campaigns:

Leverage historical sales data and forecasted trends to inform data-driven marketing campaigns. Tailor promotional activities, advertisements, and discounts based on anticipated sales patterns, such as focusing on specific product categories during peak demand periods.

Customer Segmentation and Personalization:

Utilize customer segmentation techniques to identify different customer groups with distinct purchasing behaviours. Personalize marketing efforts, product recommendations, and communication channels to target each segment effectively and enhance customer engagement.

Continuous Monitoring and Analysis:

Establish a process for continuous monitoring of sales data, market trends, and competitive insights. Regularly analyze sales performance against forecasts, conduct variance analyses, and identify factors contributing to deviations from expected trends.

Scenario Planning and Risk Mitigation:

Develop contingency plans and scenario-based forecasts to mitigate risks associated with unforeseen events or market fluctuations. Anticipate potential disruptions, such as supply chain challenges or economic shifts, and proactively adjust strategies to minimize impact.

Cross-Functional Collaboration:

Foster collaboration between sales, marketing, inventory management, and data analytics teams within Vidya Collections. Encourage cross-functional dialogue and knowledge sharing to leverage collective expertise and align strategies with sales forecasts.

Invest in Data Analytics Capabilities:

Invest in advanced data analytics tools and capabilities to enhance sales forecasting accuracy and scalability. Explore predictive analytics techniques beyond ARIMA, such as machine learning algorithms, to further improve forecasting precision.

Customer Feedback Integration:

Integrate customer feedback mechanisms, such as surveys, reviews, and social media insights, into the sales forecasting process. Incorporate qualitative data from customers to complement quantitative sales data and gain holistic insights into customer preferences and behaviour.

Continuous Improvement Culture:

Cultivate a culture of continuous improvement and learning within Vidya Collections. Encourage experimentation, feedback loops, and knowledge sharing to iterate on sales forecasting methodologies, refine strategies, and drive sustainable growth over time.

BIBLIOGRAPHY:

https://www.researchgate.net/publication/283664453_The_Implementation_of_Different_Forecasting_Techniques_for_Demand_Forecasting_in_Jute_Product_Section_in_Bangladesh

https://www.researchgate.net/publication/362909806_Prediction_and_Analysis_of_Stock_Market_using_ARIMA_Model_and_Machine_Learning_Techniques

<https://www.scirp.org/reference/referencespapers?referenceid=3413877>

https://www.researchgate.net/publication/374148601_WHEAT_MARKET_PRICE_DYNAMICS_IN_UKRAINE_QUANTITATIVE_EXPLORATION_AND_FORECASTING

https://www.researchgate.net/publication/296477650_Application_of_SARIMAX_Model_to_Forecast_Daily_Sales_in_Food_Retail_Industry

<http://icbtsproceeding.ssrui.ac.th/index.php/ICBTSSWITZERLAND/article/view/372>

<http://op.niscair.res.in/index.php/ALIS/article/view/310>

<https://iocscience.org/ejournal/index.php/mantik/article/download/2081/1640>

https://www.researchgate.net/publication/350446952_Analysis_of_Completeness_Products_and_Prices_on_the_Decision_of_Buying_ATK_Office_Stationery_at_UD_Pemancar_Ilmu_Store_in_Namlea_City

https://www.researchgate.net/publication/359712490_Consumer_Satisfaction_as_a_Mediator_on_Repurchasing_Interest_on_Price_Promotion_and_Quality_of_Service